Certified Expected-Cost Optimization for Robust Ski Rental

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We study discrete ski rental with integer purchase cost $b \geq 2$, a finite rational forecast of season length, and a prescribed expected competitive ratio at every actual integer horizon. We give an exact algorithm that minimizes forecast expected cost and returns matching primal and dual certificates on the unbounded horizon and buy-day domains. It uses a triangular prefix optimization, normalizes the policy by increasing a dual price, and recovers the robustness constraint through one moment price. An optimum needs at most $b + 1$ buy-days, including at most two days at or after b. For the uniform forecast on $\{1, \ldots, b\}$, we derive all b affine decreasing pieces of the value frontier and its constant continuation, prove uniqueness up to robustness two, and show that both support bounds are sharp. The underlying capped-geometric policy is credited to prior one-hot-forecast work. An exact example separates expected-cost optimization from a published minimum-support-level inference; a pinned software reproduction gives a third, distinct value. The algorithm has polynomial bit time in numerical b and the binary input length.

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1 Introduction

A forecast can identify the season lengths that matter most, while a competitive guarantee protects against seasons the forecast misses. The resulting design question is concrete: among all policies meeting a chosen worst-case guarantee, which one has the smallest forecast expected cost? For randomized ski rental, balancing these requirements involves the whole buying-day distribution, rather than a single threshold.

Renting costs one per day and buying costs an integer $b \geq 2$. A policy draws a positive integer buy-day T independently of the actual season length x and pays

$$C_t(x) = \begin{cases} x, & t > x, \\ b + t - 1, & t \leq x. \end{cases}$$

The input p is a finite rational probability law on positive integer horizons. Write $g(t) = \sum_x p_x C_t(x)$ and $J_p(f) = \sum_t g(t) f_t$. For rational $R \geq \rho_b := b^b/(b^b - (b-1)^b)$, we minimize $J_p(f)$ subject to

$$\mathbb{E}_f C_T(x) \leq R \min(x, b) \qquad \text{for every integer } x \geq 1. \tag{1}$$

Thus the forecast determines the objective, while actual horizons remain unbounded.

A small example explains why the choice of objective matters. At $b = 3$, $R = 3/2$, and $p = (2/5, 3/5)$ on horizons one and two, the global optimum is $39/20$. Restricting the policy to the smallest feasible level of its raw support costs instead gives $11/5$. The publisher-linked software, in six specified calls at one pinned version, returns a law costing $12/5$. Section 2 proves the first two statements and records the last as a finite software observation. This separates the optimality inference in Kim and Fan's published Appendix D.2 [KF26] from both ideal support feasibility and literal implementation behavior.

Contributions and the LP comparison. We give an explicit certifying replacement for that inference. After a finite support reduction, a backward elimination solves the prefix problem; a normalization price activates constraints in a nested order; and a moment price recovers the prescribed robustness. The algorithm returns a global lower-bound witness as well as an optimal policy. For a uniform forecast on $\{1, \ldots, b\}$, we further obtain the complete expected-cost frontier, strict-dual uniqueness, and simultaneous necessity of $b + 1$ support days and two days at or after b.

A generic rational LP already gives polynomial bit time after the finite reduction. Our contribution is the explicit construction and its structural consequences, with the same numerical-b runtime model. In particular, the moment row contains $t - 1$, which may be large when a forecast horizon is binary encoded; only the prefix block has coefficients bounded by b. Tardos's matrix-dependent arithmetic bound [Tar86] therefore does not make this whole matrix small. Arithmetic-step bounds and bit time are distinct, and we claim neither a strongly polynomial algorithm nor an asymptotic improvement over general LP.

Context and closest results. Classical rent-or-buy analysis grew from snoopy caching [KMRS88]; randomization gives the limiting ratio $e/(e-1)$ [KMMO94]. Point-prediction robustness–consistency tradeoffs and LP methods appear in Purohit, Svitkina and Kumar [PSK18] and Wei and Zhang [WZ20]. Li, Christianson and Li [LCL26] develop prediction-specific Pareto optimization, equalizing distributions and primal–dual constructions. Their inspected preprint's randomized strong-optimality statements are asymptotic in b.

The nearest model is Kim and Fan [KF26]. We use their Lemma 5.1 robustness reduction and credit their geometric and gap recurrences. The uniform policy below coincides with their Theorem C.1(ii) in Appendix C.4 one-hot-b optimizer when $R - 1 \leq 1 - 1/b$, and is another one-hot-b optimum above that cut. The uniform contribution is its expected-objective frontier and uniqueness analysis, with this underlying policy structure retained.

Distributional learning [DKT+21] combines a learned policy with a robust baseline and does provide per-instance guarantees; we optimize exactly at each supplied p, R. Besbes, Ma and Mouchtaki [BMM25] use heterogeneous data and known-radius shift to study additive regret. Cui and Dinitz [CD26b] consider unknown Wasserstein prediction error and additionally interpolate with robustness. Their benchmark is a policy knowing the true distribution. Khanafer, Kodialam and Puttaswamy [KKP13] use horizon-moment information; our moment is instead a property of the buying-day law. Tail-risk constraints [DIL+24, CD26a]

bound the probability of a large realized ratio, while here an expected-ratio cap is fixed at each horizon. Hartline, Johnsen and Shah [HJS26] study finite binary-weather sequences, with a distribution-knowing benchmark in their prior-independent analysis and a separate subgame-optimality criterion.

Set $a = b - 1$, $r = R - 1$, $K = rb$, and

$$S = \{1, \dots, b\} \cup \{d + 1 : p_d > 0,\ d \geq b\}. \tag{2}$$

Theorem 1. *For every input above, the explicit algorithm in Sections 4 and 5 returns an expected-cost minimizing robust law f, supported on S, with at most two support days $t \geq b$ and at most $b + 1$ support days in all. It returns rational $\lambda \geq 0$, $y_x \geq 0$ for $1 \leq x < b$, and $\alpha \in \mathbb{Q}$ such that*

$$g(t) + \lambda(t - 1) + \sum_{x=t}^{b-1} (b + t - 1 - x) y_x \geq \alpha \qquad (t \geq 1), \tag{3}$$

$$J_p(f) = \alpha - r \sum_{x=1}^{b-1} x y_x - \lambda K. \tag{4}$$

Empty sums are zero. Let q_R be the denominator of R, q_p the least common multiple of the probability denominators, and

$$D = b! b^b, \qquad H = q_p q_R D, \qquad L = b + \max S, \qquad Z = 2LH^4. \tag{5}$$

After computing g on S, the algorithm uses $O(\log Z)$ calls to an $O(b^2 + |S|)$ rational-operation oracle. All intermediate bit lengths and the total bit time are polynomial in numerical b and binary input length.

Figure 1 shows how the two prices separate normalization from robustness. The resulting certificates can be checked using the prefix constraints, one moment, and finitely many inequalities on S.

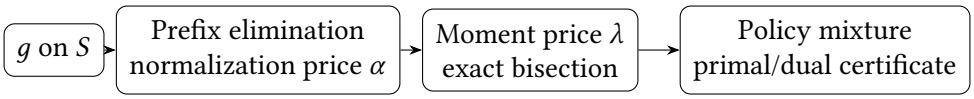


Fig. 1. The normalized prefix oracle is called at adjusted costs $g(t) + \lambda(t - 1)$; the final mixture enforces the moment cap.

2 Expected cost and raw support levels

The raw support objective is $H(f) = \sup_{t \in \operatorname{supp} f} g(t)$; on the finite set S this is a maximum. Let $h_* := \inf_{f \text{ robust}} H(f)$. Minimizing H controls every supported buy-day, whereas minimizing J_p allows a costly day if its probability is small. These objectives can therefore choose different supports.

The published Appendix D.2, printed page 30, states:

> "Therefore the optimal objective value equals the minimum feasible h, which can be found by binary search."

Here h is the raw support level, not a dual-adjusted expected objective. The following example refutes that inference in the published model [KF26]; it also applies to the corresponding arXiv v1 passage.

Proposition 2. *For $b = 3$, $R = 3/2$, $p_1 = 2/5$, $p_2 = 3/5$, the global expected-cost minimum is $39/20$, whereas $h_* = 14/5$ and the unique feasible law at this minimum support level has expected cost $11/5$.*

Proof. Direct summation gives $g = (3, 14/5, 8/5, 8/5, \ldots)$. Robustness implies $\mu(f) := \mathbb{E}(T-1) \le 3/2$, as proved in Section 3. For all $t \ge 1$, $g(t) + (7/10)(t-1) \ge 3$, with equality only at 1 and 3. Hence $J_p(f) \ge 39/20$. The law $(f_1, f_3) = (1/4, 3/4)$ attains this bound. Its costs at actual horizons 1, 2 and $x \ge 3$ are $3/2, 9/4, 9/2$, so it is robust. Below level $14/5$, all allowed days are at least 3 and have moment at least 2. At level $14/5$, $f_1 = 0$. The horizon-two constraint gives $2f_2 \le 1$, while $\mu \ge 2 - f_2$. The cap forces $f_2 = 1/2$ and all remaining mass at 3. This robust law has $J_p = 11/5$, proving the assertions. □

Table 1. Three distinct quantities for Proposition 2. The last column is the raw support objective $H(f)$ whose minimum is h_*. The software row is a finite observation; the other two rows are exact optima.

Meaning	(f_1, f_2, f_3)	J_p	μ	$H(f)$
Global expected-cost optimum	$(1/4, 0, 3/4)$	$39/20$	$3/2$	3
Minimum support-level law	$(0, 1/2, 1/2)$	$11/5$	$3/2$	$14/5$
Pinned software output	$(1/4, 3/8, 3/8)$	$12/5$	$9/8$	3

Published software observation. We executed the unchanged `SkiRentalOptimizer` class in both optimizer files supporting Figure 1 and Table 2 in the publisher-linked repository. The exact pinned version, file hashes and calls are supplied in the reproduction supplement. For constructor arguments $(3, 1.5, [1, 2], [.4, .6])$ and tolerances $10^{-5}, 10^{-9}, 10^{-12}$, all six calls returned the last law in Table 1, at levels approximately 3. Both classes rejected $14/5$ and accepted 3 in the documented feasibility calls. At level 3 the construction temporarily assigns day-three mass $9/16$, reaching internal mass $19/16$, before its existing correction leaves $3/8$. Evaluating the returned law in the original model gives moment $9/8$ and horizon costs $3/2, 3, 33/8$ at $1, 2, x \ge 3$, respectively. Thus the policy is robust and costs $12/5$. These calls concern the pinned files and tolerances; the support-level counterexample holds independently of their numerical behavior.

A different normalization changes the optimum. Dividing J_p by the fixed $\mathbb{E}_p \min(X, b)$ preserves all optimizers. In contrast, for $b = 3, R = 3/2, p_1 = p_2 = 1/2$, define $k(t) = \mathbb{E}_p[C_t(X)/\min(X, b)]$. Here $g = (3, 5/2, 3/2, \ldots)$ and $k = (9/4, 3/2, 1, \ldots)$. The global inequalities

$$g(t) + \tfrac{3}{4}(t-1) \ge 3, \qquad k(t) + \tfrac{1}{2}(t-1) \ge 2$$

are equalities only on $\{1, 3\}$ and $\{2, 3\}$, respectively. Together with $\mu \le 3/2$, they prove unique optimal laws $(f_1, f_3) = (1/4, 3/4)$ for J_p, value $15/8$, and $(f_2, f_3) = (1/2, 1/2)$ for $\mathbb{E}_f k(T)$, value $5/4$. Both laws satisfy the prefix rows. Their cross-objective values are $21/16$ and 2, respectively.

3 Robustness and finite support

Expanding the expected cost gives

$$\mathbb{E}C_T(x) = x + \sum_{t \le x} (t-1) f_t + (b - x) \sum_{t \le x} f_t.$$

For each fixed t, the cost is nondecreasing in x and converges to $b + t - 1$. Monotone convergence shows that (1) implies $\mu(f) := \mathbb{E}(T-1) \le K$. Conversely, this moment bound and the a prefix inequalities

$$\sum_{t \le x} (a + t - x) f_t \le rx, \qquad 1 \le x \le a, \tag{6}$$

imply (1): the displayed inequalities cover $x < b$, and $\mathbb{E}C_T(x) \le b + \mu(f) \le Rb$ covers $x \ge b$. This is the robustness reduction of [KF26, Lemma 5.1].

For $t \geq b$, let s be the largest member of $S \cap [b, \infty)$ not exceeding t. No positive forecast atom lies in $[s, t-1]$, so

$$g(t) - g(s) = (t - s) \Pr_{X \sim p}[X \geq t] \geq 0. \tag{7}$$

Moving mass from t to s decreases its moment, preserves every prefix constraint, and does not increase its objective. The same dominance holds for $g(t) + \lambda(t-1)$ when $\lambda \geq 0$. This elementary reduction extends the point-prediction support principle of [LCL25, Appendix E.1, arXiv v1].

The finite robust polytope on S is compact and nonempty. Indeed, the law

$$f_t = \frac{b^{t-1}(b-1)^{b-t}}{b^b - (b-1)^b}, \quad 1 \leq t \leq b,$$

sums to one, satisfies (6) with equality at $R = \rho_b$, and has moment $b(\rho_b - 1)$. These identities follow by geometric summation, or by the tight-constraint recurrence below. It remains feasible for larger R. Hence a finite-support optimum exists.

Let $\mathcal{F}$ be the simplex on S intersected with (6), omitting the moment bound. Let $\mathcal{P}$ be the polytope of nonnegative prefix vectors $(z_1, \ldots, z_a)$ satisfying (6), omitting normalization too. The next section optimizes an arbitrary linear cost on $\mathcal{F}$.

4 A normalized prefix oracle

4.1 Backward elimination and canonical schedules

For any feasible prefix through $i-1$, the next coordinate ranges over

$$0 \leq z_i \leq U_i := \frac{ri - \sum_{j<i}(a + j - i)z_j}{a}. \tag{8}$$

The numerator is $r > 0$ at $i = 1$. For $i > 1$, the preceding constraint makes it at least $r + \sum_{j<i} z_j > 0$. Setting all future coordinates to zero is feasible: increasing the horizon decreases the old left-hand side and increases the right-hand side. Thus eliminating a last coordinate creates no hidden restriction on an earlier prefix.

For arbitrary costs c_i and a scalar α, consider $\sum_i (c_i - \alpha) z_i$ on $\mathcal{P}$. At the last coordinate i, the conditional value is affine in z_i on the entire interval (8). If its current coefficient w_i is nonnegative, its minimum is at zero; if negative, it is at U_i. Substituting U_i adds the constant $w_i ri/a$ and changes each earlier coefficient w_j to $w_j - w_i(a + j - i)/a$. The remaining value is again affine on every feasible earlier prefix. Backward induction thus proves equality of the successive conditional minimum functions, and forward reconstruction attains their minimum. This proves optimality, as well as feasibility, of the elimination.

Writing $y_i = -\min(w_i, 0)/a$ gives the backward recurrence

$$y_i = \max\left\{0, \frac{\alpha - c_i - \sum_{j>i}(a + i - j)y_j}{a}\right\}. \tag{9}$$

The accumulated constant is $-r\sum_i i y_i$. Every reduced cost $c_i - \alpha + \sum_{j \geq i}(a + i - j)y_j$ is nonnegative. A positive y_i selects its tight upper row, and a positive reduced cost selects zero; therefore the reconstructed prefix attains that dual lower bound. A zero coefficient permits any point in its interval.

For $B = \{s_1 < \cdots < s_k\} \subseteq [a]$, write $v(B)$ for the prefix schedule supported on B with the prefix constraints tight at those days. With $s_0 = F_0 = 0$, subtracting consecutive tight constraints gives

$$v(B)_{s_j} = \frac{s_j - s_{j-1}}{a}(r + F_{j-1}), \qquad F_j = r\left[\prod_{\ell=1}^{j}\left(1 + \frac{s_\ell - s_{\ell-1}}{a}\right) - 1\right]. \tag{10}$$

Here F_j is the accumulated mass. The geometric and gap updates are those of [KF26, Appendix D.3–D.4]. These schedules are feasible in $\mathcal{P}$ but can have mass above one. Inserting an active day strictly increases the mass: splitting a gap $u + v$ replaces $1 + (u+v)/a$ by $(1 + u/a)(1 + v/a) = 1 + (u+v)/a + uv/a^2$; appending a day adds a factor greater than one.

4.2 Nested activation

Fix arbitrary rational costs c_t on S, including signed costs. Choose a minimizing tail day $t_\star \in S \cap [b, \infty)$, and put $c_\star = c_{t_\star}$. For $\alpha \le c_\star$, equation (9) gives the dual lower bound

$$\Phi(\alpha) = \alpha - r \sum_{i=1}^{a} i y_i \tag{11}$$

for the normalized optimization on $\mathcal{F}$. Its slope on an active interval is $1 - m(B)$, where $m(B) = \sum_i v(B)_i$: indeed $\Phi(\alpha) = \alpha + \min_{z \in \mathcal{P}} \sum_i (c_i - \alpha) z_i$. Increasing α rewards added mass. The optimum occurs when this slope crosses zero or when the cheapest tail day can fill the missing mass.

For a fixed active set B, the active y_i are affine in α; write $v_i = dy_i/d\alpha$ there and set $v_i = 0$ off B. Differentiating gives

$$a v_i + \sum_{j>i,\ j \in B} (a + i - j) v_j = 1 \quad (i \in B). \tag{12}$$

All active derivatives are positive. More generally, the reduced-slack speed

$$\delta_i = 1 - \sum_{j>i,\ j \in B} (a + i - j) v_j$$

is positive for every i. If there is no later active day, it equals one. Otherwise let k be the next active day. The equation at k implies

$$\delta_i = (k - i) \sum_{j \ge k,\ j \in B} v_j > 0. \tag{13}$$

Backward induction proves the assertion, since $v_i = \delta_i / a$ at an active day. Consequently existing positive dual coordinates increase with α, and every inactive reduced slack

$$\sigma_i = c_i - \alpha + \sum_{j>i} (a + i - j) y_j$$

decreases strictly. No day deactivates.

Algorithm 1 Normalized prefix sweep: costs c on S; output f, α, y.

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(1) Choose a cheapest tail day $t_\star \ge b$. Set $B = \varnothing$ and $\alpha = \min(c_{t_\star}, c_1, \dots, c_a)$.

(2) Add all zero-slack inactive days to B; use (9) to update y. Let v_-, v_+ be the schedules before and after this event, with masses m_-, m_+.

(3) If $m_- < 1 \le m_+$, return the zero-tail law

$$f = (1 - \eta) v_- + \eta v_+, \qquad \eta = (1 - m_-)/(m_+ - m_-). \tag{14}$$

(4) If $\alpha = c_{t_\star}$, fill $1 - m_+$ at $t_\star$ and return. Otherwise advance to the smaller of $c_{t_\star}$ and the next inactive root $\alpha + \sigma_i/\delta_i$, then repeat from step 2.

At activation, all added dual coordinates are zero, so the old dual coordinates and current slacks remain unchanged. Activating simultaneous zero slacks together fixes the convention for ties, including an activation at the tail cap. The crossing test precedes tail filling.

Lemma 3. *The sweep returns a minimizer of $\sum_{t \in S} c_t f_t$ on $\mathcal{F}$, attaining (11), with at most one tail day. It takes $O(b^2 + |S|)$ rational operations.*

Proof. At a mass crossing both v_- and v_+ minimize the unnormalized prefix problem at the activation price. Every positive old y_i corresponds to a tight constraint in both schedules, and new y_i equal zero. Their mixture (14) is prefix feasible, has mass one, and attains the normalized dual bound. When the tail cap is reached first, $v(B)$ attains the prefix bound and the remaining mass has cost exactly α, again giving equality. Positivity of the speeds and strict mass increase establish termination.

For fixed B, suffix sums of y_i, iy_i, v_i, iv_i compute all dual coordinates, slacks and speeds in $O(b)$ operations. Equation (10) reconstructs a schedule in another $O(b)$ pass. Scanning all slack roots takes $O(b)$ operations. There are at most a activations and one terminal scan. Selecting the cheapest tail takes $O(|S|)$ operations. This proves the bound. □

5 Recovering the exact moment price

Apply Lemma 3 at nonnegative price λ to

$$c_t = g(t) + \lambda(t-1), \qquad q(\lambda) = \min_{f \in \mathcal{F}} \{J_p(f) + \lambda\mu(f)\}. \tag{15}$$

If an output at zero has moment at most K, it solves the robust problem. Otherwise we maintain oracle outputs f_-, f_+ at prices $\lambda_- < \lambda_+$ with $\mu(f_-) > K \geq \mu(f_+)$. Each policy defines a line with slope $\mu(f)$ in (15). Their lower envelope is concave, so its slope decreases as the price increases. We locate the change across K, then mix the two neighboring optimal policies.

Figure 2 illustrates this local transition.

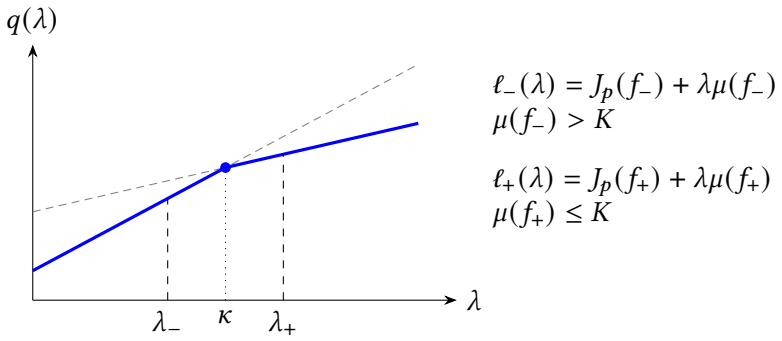


Fig. 2. Local schematic around the isolated price κ. The blue lower envelope changes from the higher-moment policy to the lower-moment policy; both are optimal at the intersection. The dashed verticals show the bracket.

Lemma 4. *With Z from (5), price $Z+1$ supplies a right endpoint. After $O(\log Z)$ bisection steps the bracket contains a common optimal price for both stored policies, recoverable exactly from their costs and moments.*

Proof. A vertex of $\mathcal{F}$ has at most b positive coordinates. Its coordinates are obtained from normalization and independent tight prefix rows, with integer coefficients of magnitude at most b. Their determinant has absolute value at most $D = b!b^b$. Cramer's rule gives a common coordinate denominator $q_R|\det|$. Its objective and moment therefore each have denominator at most H, and lie in $[0, L]$.

The intersection price of two vertex lines $J + \lambda\mu$ with distinct moments has numerator and denominator at most $Z = 2LH^4$ in reduced form: each difference has numerator at most $2LH^2$ and denominator at most H^2. Hence all positive intersections are at most Z, and distinct nonnegative intersections, including zero, are separated by at least $1/Z^2$. Beyond Z, every minimizing vertex has minimum possible moment; so does every convex combination of them. Robust feasibility gives that this minimum moment is at most K. Thus $\lambda_+ = Z + 1$ is valid.

At each bisection midpoint, replace the left endpoint if the returned moment exceeds K, and otherwise replace the right endpoint. Adding the two optimality inequalities at distinct prices proves that oracle moments are nonincreasing in price, without assuming vertex outputs or unique optima. After at most $\lceil \log_2((Z+1)Z^2) \rceil + 1$ iterations the bracket width is strictly less than $1/Z^2$.

The function q is the lower envelope of finitely many vertex lines. The strict moment bracket forces a breakpoint in the closed interval: without one, all minimizing lines have the same moment throughout it. There is at most one distinct intersection price in that interval. On each side of this price all minimizing

vertex lines have the same objective and moment. By continuity, both stored endpoint policies are optimal at the breakpoint κ. If an endpoint itself is the breakpoint, its possibly nonvertex output is already optimal there. □

The two stored moments differ, so their line intersection and a feasible mixture are

$$\kappa = \frac{J_p(f_+) - J_p(f_-)}{\mu(f_-) - \mu(f_+)}, \qquad \theta = \frac{K - \mu(f_+)}{\mu(f_-) - \mu(f_+)}, \qquad f = \theta f_- + (1-\theta) f_+. \tag{16}$$

Here $0 \le \theta < 1$, $\mu(f) = K$, and both endpoints minimize $q(\kappa)$, so their mixture does too. Recompute the prefix oracle dual at κ to obtain α, y.

Algorithm 2 Exact moment-price recovery; Z is defined in (5).

(1) Run Algorithm 1 at $\lambda = 0$. If $\mu \le K$, return this policy and its dual. Otherwise store it as f_- at $\lambda_- = 0$.
(2) Store the output at $\lambda_+ = Z + 1$ as f_+. While $\lambda_+ - \lambda_- \ge 1/Z^2$, query the midpoint price. If its moment exceeds K, replace the left endpoint; otherwise replace the right endpoint.
(3) Recover κ, θ, f by (16). Run the sweep at κ to obtain its dual and return f, κ, α, y.

5.1 Global certificates and complexity

The oracle dual at κ gives (3) on S. Equation (7) extends it to every omitted $t \ge b$; all $t < b$ already belong to S. For any robust law, taking expectations of (3) gives

$$J_p(f) \ge \alpha - r \sum_{x=1}^{b-1} x y_x - \lambda K.$$

The expectation is well defined because g is bounded, the prefix sum is finite, and robustness gives finite moment. The zero-price feasible output or the mixture (16) attains equality. Each endpoint oracle uses at most one tail day. Their mixture therefore uses at most two, proving the certificate and support assertions of Theorem 1.

Let ℓ be the binary encoding length of the input, with a sparse list of forecast atoms. Then $\log H = O(\ell + b \log b)$ and $\log Z = O(\ell + b \log b)$. Sorting forecast atoms and using prefix probability and first-moment sums computes g on S in $O(|S| + |\operatorname{supp} p| \log |\operatorname{supp} p|)$ rational operations. The simple reference program directly sums the costs and remains polynomial. Excluding preprocessing, the total number of rational operations is

$$O\big((b^2 + |S|)(\ell + b \log b)\big).$$

To bound intermediate bits, fix an active set of size $k \le a$, and let Q_c be a common denominator of the original costs in this oracle call, $M = \max(1, \max_t |c_t|)$. The triangular solve gives $y_i = u_i \alpha + v_i$, with denominators dividing a^k and $Q_c a^k$, respectively. Each substitution divides once by a, and each normalized off-diagonal coefficient has magnitude at most one. Induction on the remaining sum of absolute coefficients gives $|u_i| \le 2^k$ and $|v_i| \le 2^k M$. Every inactive slack $U\alpha + V$ therefore has the same common denominators and bounds

$$|U| \le 1 + ak2^k, \qquad |V| \le M(1 + ak2^k).$$

Its root has $O(\log Q_c + \log M + b \log b)$ bits. At each event this system is expressed in the original costs, so successive roots do not recursively enlarge their denominators. Evaluation and suffix sums add only polynomially many bits. Canonical schedules have common denominator dividing $q_R a^a$ by (10), and numerators of $O(\operatorname{len}(R) + b \log b)$ bits; normalization uses one ratio of mass differences.

After k bisections, the price denominator is at most 2^k and the numerator at most $(Z+1)2^k$, with $k = O(\log Z)$. Adjusted costs have common denominator dividing $q_p \operatorname{den}(\lambda)$ and polynomial bit length in b, ℓ. The preceding bounds apply to every oracle call. Finally (16) and the dual reconstruction use polynomial-length rationals. Standard exact rational arithmetic therefore proves polynomial total bit time, completing Theorem 1.

6 The uniform-forecast frontier

For a uniform forecast on $\{1, \dots, b\}$, every day after the forecast support has the same cost. The moment cap determines how much buying must be moved into the forecast interval; the prefix constraints limit that mass. Filling successive geometric prefix coordinates gives a complete frontier, and strict dual prices determine the policy uniquely.

Put $m = (b+1)/2$. Define z_i, d_i for $1 \le i \le b$, A_j, C_j, r_j for $0 \le j \le b$, and λ_j, s_j for $1 \le j \le b$ by

$$z_i = \frac{b^{i-1}}{a^i}, \qquad A_j = (b/a)^j - 1, \qquad C_j = (2b-j)(b/a)^j - 2b, \qquad r_j = \frac{b}{b + C_j}, \tag{17}$$

$$d_i = \frac{(b+1-i)(b+i-2)}{2b}, \qquad \lambda_j = \frac{b+j-2}{2b}, \qquad s_j = \sum_{i<j} d_i z_i - \lambda_j (b + C_{j-1}). \tag{18}$$

In particular, $A_0 = C_0 = 0$.

Theorem 5. *For the uniform forecast, $1 = r_0 > \cdots > r_b = \rho_b - 1$. If $r_j \le r \le r_{j-1}$, the unique optimal law is*

$$f_i = r z_i \ (i < j), \qquad f_j = \frac{b - r(b + C_{j-1})}{b + 1 - j}, \qquad f_{b+1} = 1 - \sum_{i \le j} f_i, \tag{19}$$

with all other masses zero. Adjacent formulas agree at shared endpoints. Its value on segment j is $m + \lambda_j b + s_j r$. For $R \ge 2$, day $b+1$ alone is optimal with value m. The value has exactly b strictly decreasing affine pieces followed by this plateau. The shared frontier representation, a single policy/value query, and a dual query each take $O(b)$ rational operations; every rational uses $O(b \log b + \mathrm{len}(R))$ bits. For $r_b < r < r_{b-1}$, all $b+1$ masses on $\{1, \dots, b+1\}$ are positive. Thus both support bounds of Theorem 1 are simultaneously sharp.

The capped-geometric law in (19) agrees with the known one-hot-b optimizer in the range described in Section 1. The plateau's unconstrained forecast behavior also appears in the discrete Bayesian model of Kang, Park and Fan [KPF25]. We prove the frontier and uniqueness using its forecast-specific dual.

Proof. Direct summation gives $g(i) = m + d_i$ for $i \le b$, and $g(t) = m$ for $t \ge b+1$. Geometric summation, or subtraction of consecutive expressions, gives

$$A_j = \sum_{i \le j} z_i, \quad C_j = \sum_{i \le j} (b+1-i) z_i, \quad M_j := \sum_{i \le j} (i-1) z_i = b - (b-j)(b/a)^j.$$

Thus C_j increases strictly, $r_b = a^b/(b^b - a^b)$, and $M_j \le b$, strictly for $j < b$. At $r = r_j$, the policy has $f_i = r_j z_i$ for $i \le j$; at $r = r_{j-1}$, $f_j = 0$. These endpoint identities prove agreement and, by affinity, $0 \le f_j \le r z_j$ throughout the interval.

The full geometric prefix makes its rows tight, since $a f_i = r + \sum_{k<i} f_k$. The displayed policy has the first $j-1$ rows tight and the jth row feasible when $j < b$. With no further early mass, subsequent left sides decrease while their right sides increase, so all prefix rows hold. At endpoint r_j the residual tail is $1 - r_j A_j \ge 0$: using $C_j = bA_j - M_j$, this inequality is equivalent to $M_j \le b$. The other endpoint is the preceding policy, so affinity proves tail nonnegativity on the whole segment. Finally,

$$\mu = b - \sum_{i \le j} (b+1-i) f_i = rb. \tag{20}$$

This proves feasibility, including the classical endpoint where the tail vanishes. For $r \ge 1$, day $b+1$ has moment $b \le rb$ and cost m.

For segment j, choose $\lambda = \lambda_j > 0$, $\alpha = m + \lambda b$, and $y_i = 0$ for $i \ge j$. Choose y_i backward for $i < j$ to make (3) an equality at those days. It is already an equality at j, since $d_j = \lambda(b+1-j)$. For $j \ge 2$, the last early price satisfies $a y_{j-1} = (b - j + 2)/(2b) > 0$. Subtracting consecutive equality equations gives

$$a y_i - \sum_{k=i+1}^{j-1} y_k = \frac{b + j - 2i}{2b} > 0 \qquad (1 \le i < j - 1). \tag{21}$$

Hence all y_i for $i < j$ are positive. For $j < t \le b$, the reduced cost is $(b+1-t)(t-j)/(2b) > 0$; at $b+1$ it is zero, and for $t > b+1$ it is $\lambda(t-b-1) > 0$. This certifies the inequality for every buy-day. The displayed policy uses only equality days, tightens every positively priced row, and satisfies (20), proving optimality.

Conversely, equality in the dual bound forces every optimum to have support in $\{1, \dots, j, b+1\}$, its first $j-1$ prefix rows tight, and moment rb. These triangular equations and normalization uniquely give (19), including shared endpoints and $r = 1$. Substitution into $J_p = m + \sum_i d_i f_i$ gives the value claimed. Its slopes satisfy

$$s_1 = -(b-1)/2 < 0, \qquad s_{j+1} - s_j = -\frac{b + C_j}{2b} < 0.$$

As r increases the segments are traversed in reverse j order, so all b distinct negative slopes occur, followed by zero. In the lowest open segment ($j = b$), the first b masses are positive. The tail is affine, zero at r_b and positive at r_{b-1}, so it too is positive inside. Uniqueness excludes every smaller-support optimum, even on later days.

Compute successive z_i and accumulate $A, C, \sum d_i z_i$ to store one slope, intercept and breakpoint per j. This takes $O(b)$ operations and shares the prefix data across segments. An interval scan, policy construction and backward dual solve each take $O(b)$ operations. Powers b^j, a^j have $O(b \log b)$ bits. The sums share denominator a^b, the d_i have denominator $2b$, and the dual recurrence has depth at most b with diagonal a. All resulting rationals have $O(b \log b + \mathrm{len}(R))$ bits. This bound concerns the compact representation and one query, rather than printing b dense policy vectors. □

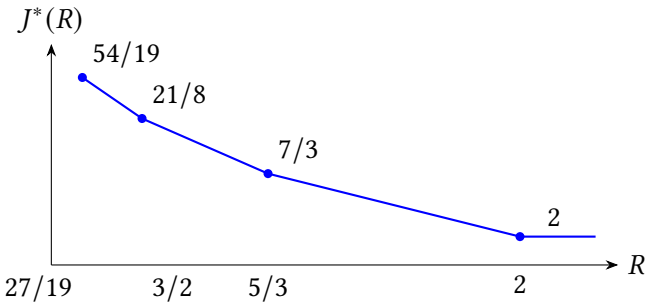


Fig. 3. The exact uniform frontier at $b = 3$: three affine pieces and the plateau. On $27/19 < R < 3/2$, the unique optimum needs all four buy-days.

Further questions. Further directions include worst-case bounds on frontier size, direct structural formulas for arbitrary forecasts, and improved dependence on numerical b. The general-forecast structural frontier is left for future work.

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